

Token-World: Compact VLM Representations for Policy-Oriented World Modeling

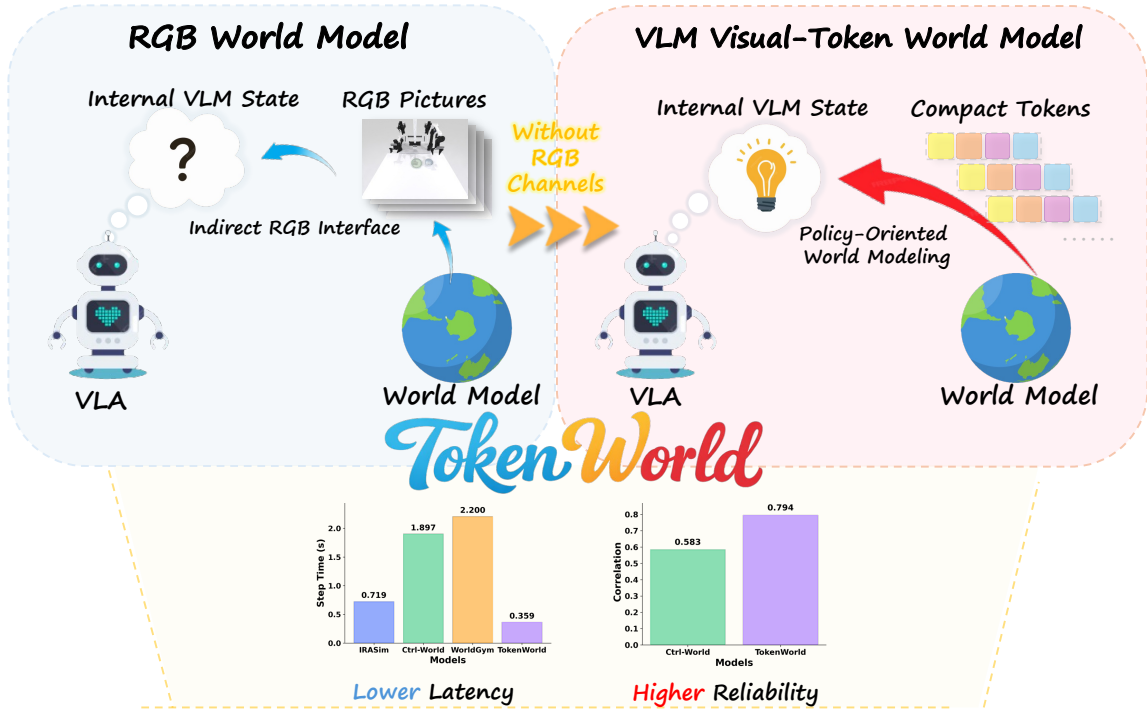


Fig. 1: **Token-World simulates directly in compact VLM visual-token space.** Unlike RGB-based world models that generate future images and re-encode them into policy inputs, Token-World models action-conditioned dynamics directly in compact VLM tokens. This design avoids intermediate RGB generation, reduces simulator step time to 0.359 s, and yields more reliable closed-loop policy evaluation ($r = 0.794$ vs. 0.583 for Ctrl-World).

Abstract—World-model simulators for vision-language-action (VLA) systems often predict future RGB frames and then re-encode them into policy inputs, introducing an indirect interface between simulation and downstream policy execution. We instead investigate whether world dynamics can be modeled in a compact, policy-oriented state derived from VLM visual tokens. A key challenge is that raw VLM visual tokens are high-dimensional and difficult to model autoregressively. To address this, we introduce Token-World, an action-conditioned world model that compresses VLM features into a compact token state, learns future dynamics in this reduced space, and maps predicted states back to the original policy-facing representation for downstream use. Across manipulation benchmarks, Token-World improves open-loop feature fidelity and policy-action consistency over recent world-model simulators, while degrading more slowly over long horizons. In closed-loop evaluation, it better tracks reference policy performance than Ctrl-World ($r = 0.794$ vs. 0.583) with higher simulation efficiency. Ablations further show that compact-representation design and dimensionality substantially affect future-state prediction.

I. INTRODUCTION

World models [1] offer a promising route toward scalable embodied learning by serving as learned simulators of action-conditioned environment dynamics [2], [3], [4], [5]. Instead of executing every behavior in the physical world, an agent can roll out future observations inside the model, which is especially valuable in robotics where real-world interaction is costly and slow[5]. Recent work has therefore explored world models as simulators from several perspectives: using predicted futures to evaluate policies [6], [7], synthesizing additional interaction data for policy learning[8], [9], and performing reinforcement learning through imagined rollouts[10], [11], [12], [13].

As world models are increasingly used as learned simulators, they are often paired with VLA policies that map visual observations and language instructions to actions [14], [15], [16], [17]. However, existing pipelines often use an indirect simulation interface, where future RGB

observations are first predicted by the world model, then re-encoded into VLA visual tokens before being consumed by the VLA policy [6], [8], [10], [11], [12], [13]. As world models move toward scalable simulators for VLA agents, this indirect interface becomes a critical bottleneck: the simulator predicts human-viewable pixels, whereas the downstream policy ultimately consumes policy-facing visual tokens. This mismatch is not merely computational: RGB reconstruction encourages the simulator to allocate capacity to visual details that may be irrelevant to the policy, while token-space rollout directly targets the representation used for downstream action generation.

A representation-aligned alternative is to simulate the future directly in the VLM visual-token space consumed by the policy. Such an interface avoids repeatedly reconstructing RGB observations and re-encoding them into policy inputs during imagined rollouts. However, policy-facing VLM tokens are high-dimensional representations optimized primarily for perception and action rather than generative dynamics modeling [15], [16], [17]. Directly predicting their temporal evolution therefore imposes both a substantial modeling burden and considerable computational cost.

In this work, we introduce **Token-World**, an autoregressive action-conditioned world-model simulator that rolls out in compact policy-aligned VLM token space. Its dynamics model is a flow-matching DiT, without relying on pretrained video-generation models. To make high-dimensional policy-facing VLM features tractable for dynamics learning, Token-World compresses the feature dimension into a compact semantic token state. Future states are predicted in this compact space and mapped back to the original VLM visual-token space only when consumed by the downstream policy, avoiding intermediate RGB generation throughout the rollout.

Across manipulation benchmarks, Token-World improves open-loop feature fidelity and policy-action consistency over recent world-model simulators, while degrading more slowly over long rollout horizons. In closed-loop evaluation, its simulated success rates track reference policy performance more closely than Ctrl-World ($r = 0.794$ vs. 0.583), while providing higher simulation efficiency. Controlled ablations further show that both the design and dimensionality of the compact VLM representation substantially affect future-state prediction. In summary, our contributions are threefold:

- We study direct VLM visual-token simulation as an alternative to RGB-based world-model pipelines, enabling the simulator to operate on the representation interface consumed by downstream VLA policies.
- We present Token-World, an autoregressive action-conditioned world-model simulator that uses a flow-matching DiT to model dynamics in a compact VLM visual-token space, without relying on pretrained video-generation models.
- We validate Token-World through open-loop prediction, closed-loop policy evaluation, and representation ablations, showing improved long-horizon feature and policy-action fidelity, more reliable policy success estimation, lower simulation cost, and the importance of compact-

representation design for dynamics prediction.

II. RELATED WORK

A. World Models as Learned Simulators

World models learn action-conditioned dynamics to support planning and policy learning through imagined rollouts [1], [2], [3], [4], [5]. Recent robotic world models have been used for policy evaluation [18], [6], synthetic interaction generation [8], [19], and policy optimization [10]. Token-World focuses on a complementary question: what representation should serve as the state of a world model coupled with a VLA policy?

B. Representation Design for Latent World Models

Latent world models predict future states in learned representation spaces. DINO-WM and related methods model pretrained visual features [20], [21], [22], while JEPA-style models predict future latent embeddings [23], [24]. Beyond generic feature latents, Mask World Model predicts future semantic masks as a structured geometric bottleneck that suppresses task-irrelevant appearance variation [25]. Recent VLA-oriented methods move prediction closer to policy representations: DIAL predicts VLM features [26], AHEAD forecasts future VLA patch tokens [27], and LaWAM predicts latent visual subgoals [28]. Together, these works highlight the importance of the predictive representation; our focus is instead on how a compact dynamics state should be constructed from high-dimensional policy-facing representations for action-conditioned world modeling.

Related generative models have also studied representation design for diffusion. Representation Autoencoders use pretrained semantic features as diffusion latents [29], while PS-VAE maps high-dimensional representation features into a compact, KL-regularized latent space for image generation and editing [30]. Compression has also appeared in world models: DeltaWorld compresses DINO feature changes into a delta token [31], and OneWM-VLA compresses each visual view into a semantic token [32]. Token-World instead compresses policy-facing VLM tokens along the feature dimension while preserving their spatial layout, distinguishing it from approaches that aggregate frame- or view-level information into a small number of compact tokens.

III. TOKEN-WORLD

A. Overview

Token-World separates the representation consumed by the policy from the state space modeled by the world model. Given policy-facing VLM tokens $z_t \in \mathbb{R}^{N \times D_0}$, we first map each token into a compact representation

$$c_t = C_\phi(z_t) \in \mathbb{R}^{N \times d}, \quad d \ll D_0, \quad (1)$$

while preserving the number and spatial arrangement of tokens. Token-World then learns action-conditioned dynamics in this compact state space. Predicted compact states can be mapped back through a frozen decoder,

$$\hat{z}_t = G_\phi(\hat{c}_t), \quad (2)$$

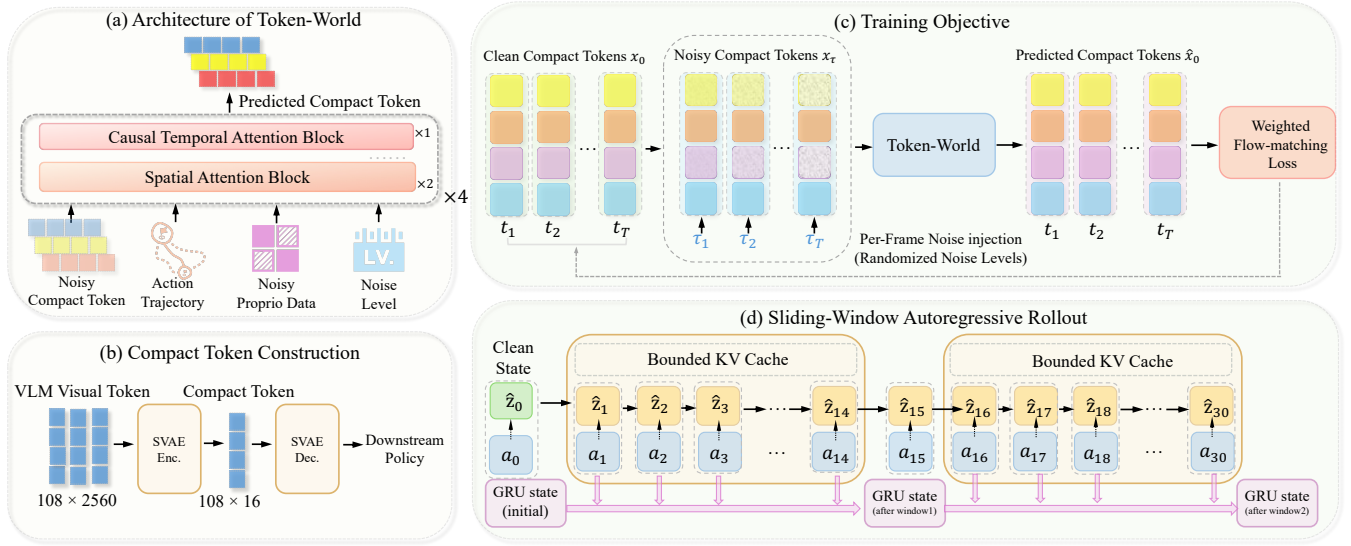


Fig. 2: **Token-World overview.** (a) Token-World uses a flow-matching DiT to model action-conditioned dynamics directly in compact token space. (b) VLM visual tokens are compressed along the feature dimension into a compact token state and decoded back only when queried by the downstream policy. (c) The dynamics model is trained with an x_0 -parameterized flow-matching objective under diffusion forcing, using independently randomized noise levels across temporal states. (d) Compact states are rolled out autoregressively with a sliding temporal window for long-horizon simulation.

when policy-facing features are required for downstream control or evaluation.

The dynamics model is implemented as a spatio-temporal Transformer with factorized spatial and causal temporal attention [33], [34], together with recurrent temporal memory for context beyond the finite attention window. We train the transition model using an x_0 -parameterized flow objective with shortcut forcing [35], [36], [37], and perform autoregressive rollout directly in the compact state space.

B. Compact VLM World State

We use a separately trained semantic VAE (S-VAE) to construct the compact state space. The frozen VLM encoder produces $z_t \in \mathbb{R}^{N \times D_0}$, and the compression encoder parameterizes a diagonal Gaussian posterior

$$q_\phi(c_t | z_t) = \mathcal{N}(\mu_\phi(z_t), \text{diag} \sigma_\phi^2(z_t)). \quad (3)$$

During codec training, compact states are sampled using

$$c_t = \mu_\phi(z_t) + \sigma_\phi(z_t) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (4)$$

while the posterior mean is used when exporting representations for world-model training.

The decoder reconstructs the original policy-facing representation as

$$\tilde{z}_t = G_\phi(c_t). \quad (5)$$

Importantly, compression is applied along the feature dimension only: no spatial pooling or token fusion is performed, so the original spatial token layout is retained. In our main configuration, the VLM representation contains $N = 108$ spatial tokens with $D_0 = 2560$ channels, which are compressed to $d = 16$ channels per token.

The codec is optimized independently of the dynamics model using

$$\mathcal{L}_{\text{codec}} = \mathcal{L}_{\text{MSE}}(\tilde{z}_t, z_t) + \mathcal{L}_{\text{cos}}(\tilde{z}_t, z_t) + \lambda_{\text{KL}} D_{\text{KL}}(q_\phi(c_t | z_t) \| \mathcal{N}(0, I)). \quad (6)$$

where λ_{KL} controls the latent regularization. After training, both C_ϕ and G_ϕ are frozen when learning world dynamics.

C. Action-Conditioned Dynamics Model

We define the compact world state as

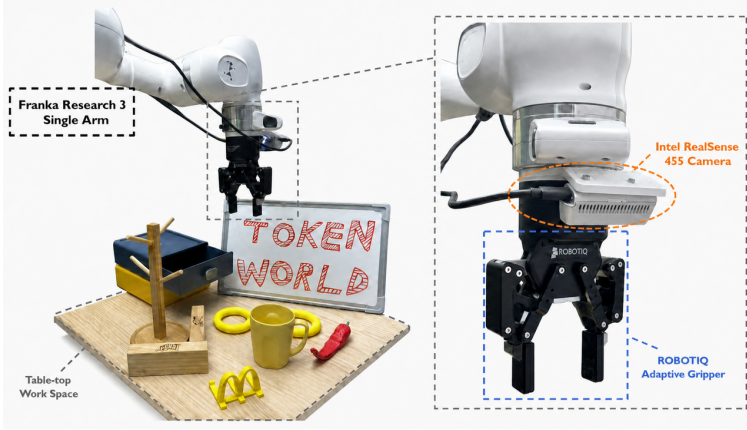
$$x_t = (c_t, p_t), \quad (7)$$

where p_t denotes the robot proprioceptive state. Token-World models the action-conditioned transition

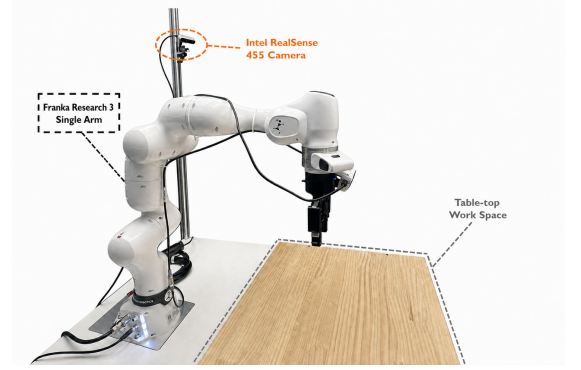
$$p_\theta(x_{t+1} | x_{\leq t}, a_t). \quad (8)$$

Compact visual tokens, proprioception, and actions are projected into a shared model space and processed by a spatio-temporal Transformer. Spatial attention captures interactions among tokens within each timestep, while causal temporal attention models their evolution over time. We factorize the two attention axes to avoid dense attention over the full space-time sequence [34].

Because temporal attention operates over a finite context window, we additionally maintain a fixed-size GRU state [38] across temporal blocks. This recurrent state summarizes earlier interaction context that has fallen outside the active attention window and is propagated across successive rollout windows. We treat the Transformer and recurrent state as components of the transition model rather than as separate representation spaces.



(a) Table-top manipulation platform and wrist-mounted sensing setup.



(b) External view of the real-world data-collection setup.

Fig. 3: Real-world evaluation setup. We collect manipulation trajectories using a Franka Research 3 arm equipped with a Robotiq adaptive gripper and two Intel RealSense 455 cameras. The real-world benchmark contains six manipulation tasks (hang on M, hang on cup, stack jenga, stack ring, put jenga in drawer, put chili in drawer) with 100 trajectories per task.

D. Training and Autoregressive Rollout

a) *Flow-matching objective.*: Following [35], [39], we train the dynamics model using an x_0 -parameterized flow objective. Let x denote a clean future compact world state and let $\epsilon \sim \mathcal{N}(0, I)$. For flow time $\tau \in [0, 1)$, we construct

$$x_\tau = (1 - \tau)\epsilon + \tau x. \quad (9)$$

The transition model predicts the clean endpoint conditioned on the available interaction history,

$$\hat{x} = f_\theta(x_\tau, \tau, \mathcal{C}), \quad (10)$$

where $\mathcal{C}$ contains previous compact states, proprioception, actions, and recurrent context.

The endpoint prediction induces the flow

$$v_\theta = \frac{\hat{x} - x_\tau}{1 - \tau}, \quad (11)$$

which is trained against the target flow using

$$\mathcal{L}_{\text{flow}} = \mathbb{E} \left[w(\tau) \|v_\theta - v^*\|_2^2 \right], \quad w(\tau) = 0.9\tau + 0.1. \quad (12)$$

We further adopt shortcut forcing [35], [36], [37], which trains the model on intermediate states generated from its own predictions and supports efficient few-step generation. The VLM encoder and semantic VAE remain frozen throughout world-model training.

b) *Autoregressive rollout.*: At inference time, the observed compact state initializes the model context. Future states are generated sequentially by denoising an initialized noisy state conditioned on the current history and action. Each denoised prediction is appended to the rollout history and used to condition subsequent transitions,

$$\hat{x}_{t+k+1} \sim p_\theta(\cdot \mid \hat{x}_{\leq t+k}, a_{t+k}). \quad (13)$$

Temporal attention is restricted to a bounded context window, while the recurrent state is propagated across windows to retain longer-term interaction context.

The rollout remains entirely in the compact state space. When policy-facing features are required, the predicted compact visual state is mapped back through the frozen semantic decoder,

$$\hat{z}_{t+k} = G_\phi(\hat{c}_{t+k}), \quad (14)$$

and the reconstructed VLM representation is consumed directly by the downstream policy. No RGB reconstruction is involved in the autoregressive dynamics or policy-feedback loop.

c) *RGB rendering for evaluation.*: For qualitative visualization and manual task-success assessment, a separately trained and frozen convolutional decoder D_{rgb} reconstructs 224×224 RGB frames from predicted VLM visual-tokens. This decoder is used only for rendering and does not participate in world-model rollout or policy inference.

IV. EXPERIMENTS

We evaluate Token-World from three complementary perspectives: open-loop prediction fidelity, closed-loop policy evaluation, and representation design. We first compare Token-World with recent robotic world models and analyze how prediction quality changes over long rollout horizons. We then test whether these gains translate into more reliable and efficient closed-loop policy simulation by comparing simulated and reference success rates. Finally, controlled ablations study how the design and dimensionality of the compact S-VAE state affect dynamics prediction.

A. Experimental Setup

a) *Datasets.*: We evaluate Token-World on both simulated and real-world manipulation data. For simulation, we use 50 manipulation tasks from RoboTwin [40], with 100 trajectories per task, including 50 successful and 50 failed demonstrations, for a total of 5,000 trajectories. For real-world evaluation, we collect a Franka manipulation dataset containing six tasks and 600 successful trajectories. Both

TABLE I: **Open-loop world-model fidelity on simulated and real-world manipulation.** Feature metrics compare predicted and ground-truth frozen VLM representations. Policy-action metrics compare the outputs of the same frozen policy conditioned on predicted and ground-truth representations. For RGB-generative baselines, generated frames are re-encoded by the frozen VLM before evaluation. Higher cosine similarity and lower NMSE are better.

Method	Venue	RoboTwin (50 Tasks)				Real Franka (6 Tasks)			
		VLM Feature Fidelity		Policy-Action Consistency		VLM Feature Fidelity		Policy-Action Consistency	
		Cosine $\uparrow$	NMSE $\downarrow$	Cosine $\uparrow$	NMSE $\downarrow$	Cosine $\uparrow$	NMSE $\downarrow$	Cosine $\uparrow$	NMSE $\downarrow$
IRASim	ICCV'25	0.7372	0.5685	0.9260	0.1533	0.7061	0.6095	0.9174	0.1702
Ctrl-World	ICLR'26	0.7097	0.6437	0.9265	0.1463	0.7213	0.5719	0.9288	0.1410
WorldGym	ICLR'26	0.7146	0.6157	0.9251	0.1602	0.6697	0.6930	0.9089	0.1981
Token-World	Ours	0.7714	0.4892	0.9439	0.1145	0.7506	0.5129	0.9402	0.1208

datasets use a 9:1 train-validation split, and all compared methods share the same training and evaluation data.

b) Baselines.: We compare Token-World with three recent robotic world models: IRASim [41] (ICCV 2025), Ctrl-World [8] (ICLR 2026), and WorldGym [6] (ICLR 2026). These methods represent competitive learned simulators based on generative visual dynamics.

c) Open-loop evaluation.: Under open-loop action replay, each world model predicts future states from the recorded action sequence. We evaluate VLM-feature fidelity using NMSE and cosine similarity. To measure policy-action consistency, we feed the predicted and ground-truth representations separately to the same frozen policy and compare the resulting actions. Metrics are averaged over trajectories within each task and then macro-averaged across tasks.

d) Closed-loop evaluation and efficiency.: For closed-loop policy evaluation, we use three StarVLA variants (StarVLA-OFT, StarVLA-PIv3, and StarVLA-GR00T) [42], which share the same Qwen3-VL [43] backbone but use different action heads. Each rollout is initialized from the same recorded state as its reference environment counterpart, after which predicted states are fed back to the policy to generate subsequent actions. We compare simulated and reference success rates using Pearson correlation, with success manually assessed from decoded RGB rollouts.

We measure batch-1 latency on PPU-ZW810E accelerators using the same inference settings as the fidelity evaluations: RGB baselines use 50 denoising steps, whereas Token-World uses 16 steps with shortcut forcing. Latency is reported per predicted observation to account for different numbers of future observations generated per diffusion call. Timing includes RGB decoding and VLM re-encoding for RGB baselines, and compact-state prediction plus direct decoding to policy-facing VLM features for Token-World.

B. Open-Loop Prediction Fidelity

We first evaluate Token-World under open-loop action replay, considering both overall prediction fidelity and long-horizon degradation.

a) Overall open-loop fidelity.: Table I compares Token-World with recent robotic world models. On RoboTwin,

Token-World achieves a feature cosine similarity of 0.7714 and NMSE of 0.4892, compared with 0.7372/0.5685 for IRASim and 0.7097/0.6437 for Ctrl-World. The gains persist after policy inference, with an action cosine similarity of 0.9439 and NMSE of 0.1145, compared with 0.9260/0.1533 and 0.9265/0.1463, respectively.

b) Long-horizon rollout fidelity.: We compare Token-World and Ctrl-World at replay chunks 0, 5, 10, and 15, with 16 actions per chunk. Figure 5 reports the corresponding chunk-wise Qwen-feature and action similarity.

c) Qualitative long-horizon rollout.: Figure 4 provides a qualitative comparison over a long open-loop rollout. Token-World better preserves task-relevant object configurations and robot-object interactions across the trajectory, while Ctrl-World gradually deviates from the reference evolution at later steps. For visualization, Token-World predictions are decoded from the compact token space back to RGB; RGB reconstruction is not used during the rollout itself.

C. Closed-Loop Policy Evaluation

Open-loop evaluation measures prediction under fixed action sequences, while closed-loop simulation tests whether these predictions remain useful once they influence subsequent policy actions. We therefore evaluate Token-World as a policy simulator by measuring whether it preserves task-level outcomes under closed-loop rollout.

a) Success-rate correlation.: Figure 6 compares simulated and reference success rates across policy-task pairs. Token-World increases the Pearson correlation from 0.583 to 0.794 over Ctrl-World, with a regression closer to the oracle $y = x$. This indicates that its open-loop fidelity gains translate into more reliable closed-loop policy evaluation.

b) Simulation efficiency.: As shown in Fig. 6, Token-World achieves the lowest per-step simulation time at 0.359 s, yielding 2.0 $\times$, 5.3 $\times$, and 6.1 $\times$ speedups over IRASim, Ctrl-World, and WorldGym, respectively.

D. Representation Design Ablations

We further investigate the source of these gains through dimensionality studies, dynamics validation, and a controlled S-VAE versus VAE comparison.

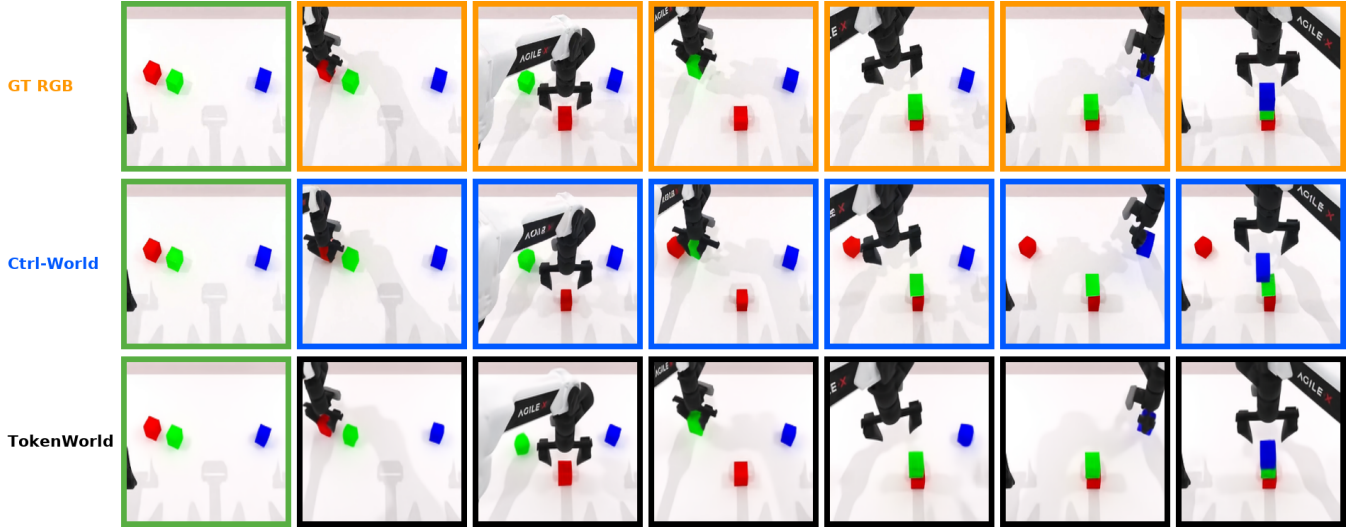


Fig. 4: **Qualitative comparison of long-horizon open-loop rollout.** We visualize seven states along the same action-replay trajectory. The top row shows the reference RGB observations, followed by predictions from Ctrl-World and Token-World. Token-World more closely preserves the task-relevant object configuration and interaction progression over long horizons. Token-World predictions are rendered to RGB using the auxiliary decoder only for visualization.

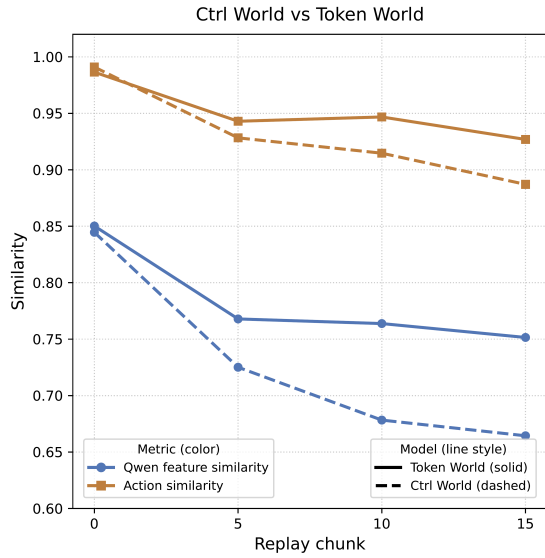


Fig. 5: **Long-horizon open-loop fidelity under chunk-based autoregressive rollout.** Results are reported at replay chunks 0, 5, 10, and 15, with 16 steps a chunk. Each point denotes the chunk-wise average VLM-feature and action similarity. Token-World degrades more slowly than Ctrl-World over long-horizon rollout.

a) Reconstruction-modelability trade-off. We first vary the channel dimension of the S-VAE while preserving the same spatial token structure. For each representation, we measure both reconstruction fidelity and diffusion-modelability properties.

Following prior analyses of latent diffusability, we use Spectral Energy Concentration (SEC) [44] and Local-vs-Distant Similarity (LDS) [45] to characterize spectral smoothness and

TABLE II: **Reconstruction and modelability of S-VAE representations across compact-state dimensions.** Increasing the latent dimension improves reconstruction fidelity, whereas the modelability metrics exhibit the opposite trend.

d	NMSE↓	Cos.↑	SEC↓	LDS↑
8	0.1026	0.9389	0.3511	0.3100
16	0.0791	0.9528	0.3854	0.2534
24	0.0694	0.9587	0.3960	0.2383
32	0.0633	0.9623	0.4058	0.2274
48	0.0562	0.9668	0.4269	0.1969
96	0.0445	0.9739	0.4394	0.1595
128	0.0409	0.9762	0.4740	0.1182

spatial structure, respectively. Lower SEC indicates smoother latent features, while higher LDS indicates stronger local spatial structure. As shown in Table II, increasing the compact dimension from 8 to 128 improves reconstruction (NMSE: 0.1026 $\rightarrow$ 0.0409; Cos.: 0.9389 $\rightarrow$ 0.9762), but increases SEC (0.3511 $\rightarrow$ 0.4740) and decreases LDS (0.3100 $\rightarrow$ 0.1182). This suggests that higher reconstruction fidelity does not necessarily correspond to more diffusion-friendly latent properties.

b) Effect of compact-state dimension on dynamics prediction. Representation-level diagnostics alone do not establish whether the observed trend matters for an actual world model. We therefore train the same dynamics backbone using S-VAE representations with different compact dimensions. The architecture, training data, optimization protocol, and evaluation procedure are held fixed; only the compact-state dimensionality is varied.

Among the evaluated dimensions, $d = 16$ provides the best overall trade-off for dynamics prediction. It achieves the lowest feature NMSE (0.1517), highest feature cosine simi-

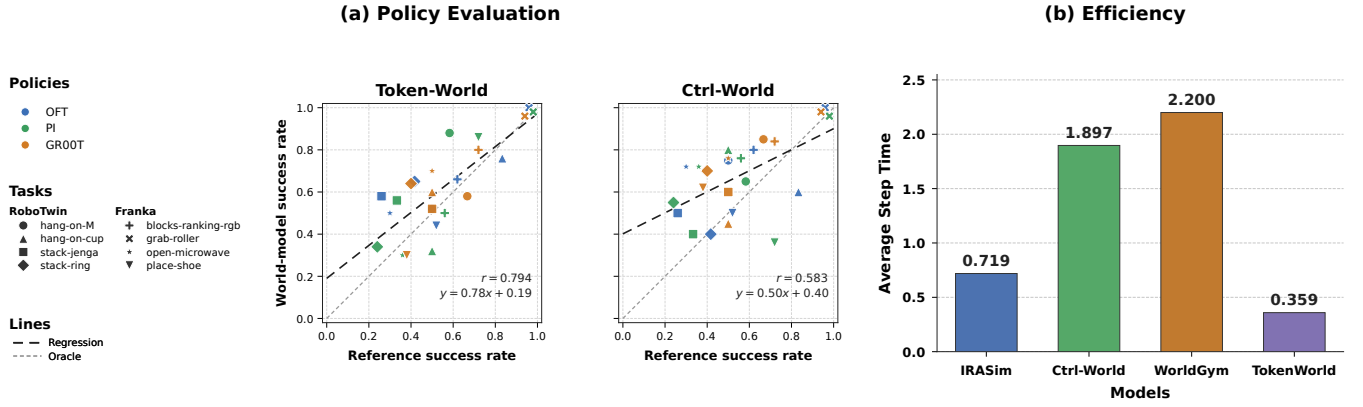


Fig. 6: **Policy evaluation fidelity and simulation efficiency.** (a) Correlation between policy success rates measured in learned world models and the corresponding reference environments. Each point represents one policy–task pair. The gray dotted line indicates the oracle relation $y = x$, and the black dashed line denotes linear regression. (b) Average per-step simulation time of different world-model simulators. Lower is better.

TABLE III: **Effect of S-VAE dimensionality on downstream dynamics prediction.** The dynamics backbone, training data, and optimization protocol are fixed; only the compact-state dimension is varied.

Representation	Feat. NMSE↓	Feat. Cos.↑	Act. NMSE↓	Act. Cos.↑
S-VAE-8	0.1549	0.8254	0.0259	0.9603
S-VAE-16	0.1517	0.8306	0.0247	0.9641
S-VAE-32	0.1634	0.8182	0.0275	0.9596
S-VAE-48	0.1698	0.8109	0.0236	0.9638

TABLE IV: **Controlled representation ablation in Token-World.** Both variants use the same dynamics architecture and 100K-step training budget; only the representation is changed.

Representation	Feat. NMSE↓	Feat. Cos.↑	Act. NMSE↓	Act. Cos.↑
SDXL-VAE	0.8164	0.621	0.2586	0.875
S-VAE	0.4892	0.7714	0.1145	0.9439

larity (0.8306), and highest action cosine similarity (0.9641). Reducing the dimension further to 8 slightly degrades both feature and action consistency, while larger dimensions 32 and 48 yield worse future-feature prediction. These results indicate that dynamics performance is non-monotonic with representation capacity, supporting $d = 16$ as the operating point for Token-World.

c) *S-VAE versus SDXL VAE representations.*: To isolate the effect of world-state representation, we train the same Token-World dynamics model using either S-VAE features or the VAE latents of Stable Diffusion XL (SDXL) [46], with the same 100K-step training budget. S-VAE substantially outperforms the SDXL VAE, reducing feature NMSE from 0.8164 to 0.4892 and action NMSE from 0.2586 to 0.1145, while improving the corresponding cosine similarities from 0.621/0.875 to 0.7714/0.9439.

This controlled ablation shows that, with the dynamics model fixed, the choice of world-state representation substantially affects future-state prediction. Together with the

dimensionality ablations, these results suggest that effective dynamics modeling depends not only on compression level, but also on the structure of the representation.

V. CONCLUSION

We presented Token-World, an autoregressive world-model simulator that learns action-conditioned dynamics in a compact policy-facing VLM token space. By compressing high-dimensional VLM features into a dynamics-friendly state, Token-World avoids intermediate RGB generation while retaining compatibility with downstream VLA policies. Experiments on simulated and real-world manipulation show improved open-loop feature and policy-action fidelity, more stable long-horizon rollouts, stronger agreement with reference policy performance in closed-loop evaluation, and lower simulation latency than recent world-model baselines. Representation ablations further highlight the importance of compact-state design for dynamics modeling. Our current evaluation focuses on manipulation tasks and policies sharing a common VLM backbone; extending Token-World across policy backbones and to more diverse interaction data remains important future work.

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